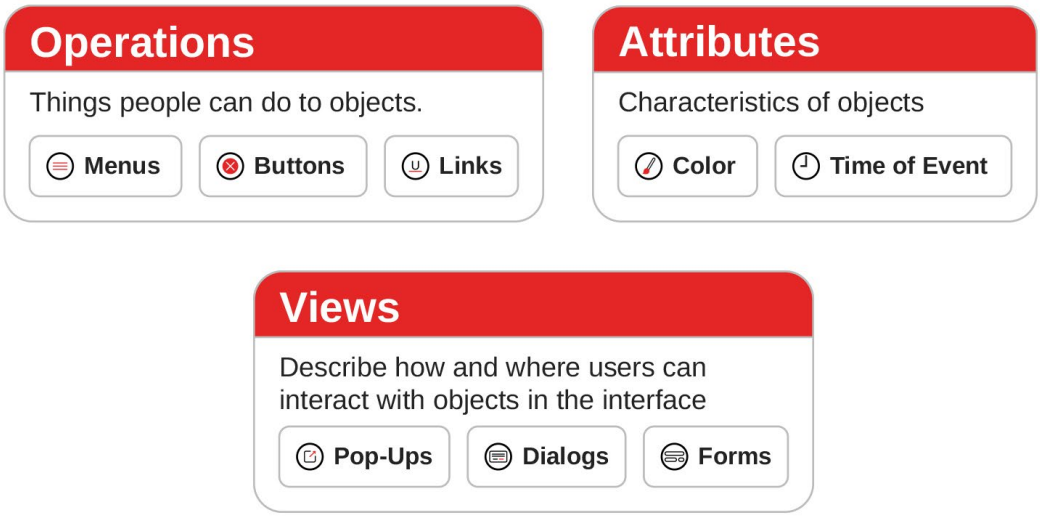

Master the Language of Object-Orientation

Object-Oriented UI Design Glossary

As you adopt the object-oriented UI (user interface) design approach, one of your greatest tools is the language. Speaking “Object Orientation (OO)” helps you think in systems and identify relationships, behaviors, and structures the way developers do. That shared mindset also makes collaboration with developers smoother since you’ll be communicating with a shared language, rather than translating across disciplines; that’s a career superpower.

How To Use This Template

Think of this glossary as your go-to phrasebook for the world of object-oriented UI design. Don’t worry, it’s not about memorizing jargon. The beauty of OO language is that, like the approach itself, it mirrors how we naturally think and label things in everyday life.

At the end, you’ll find space to write new terms as you discover them, and make this glossary your own personal reference that grows with your understanding.

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Key Object-Oriented UI Design Terms

Object

A “thing” users care about (real or conceptual) that bundles data (attributes) and behavior (operations). It mirrors how people think and how the UI should explain itself.

Class

The blueprint for objects (e.g., Hotel, Reservation). A class can produce many object instances.

Instance

A specific object created from a class (e.g., reservation #1234).

View

A UI surface that lets users see/edit object data and trigger actions (e.g., <<View>> HotelCheckinView).

Attributes (UI fields)

The data an object or a view holds/displays (e.g., departureDate, nights). In UI models, attributes often map directly to fields; some are display-only, and some are temporary and disappear when the view closes.

Actions / Operations (Methods)

Things users can do (e.g., checkIn(), viewRooms()).

State

A condition an object/view can be in (e.g., a call button being “pushed”).

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Object Relationships

Conceptual model

For UI designers, this shows the important concepts and relationships the UI must teach users. For engineers, this shows the inner workings of the system.

Association

A relationship between objects (e.g., Guest *stays* in Room). Can be named/directed.

Association class

A relationship turned into a class because it carries its own data (e.g., Reservation, Employment, Marriage).

Aggregation (“has-a”)

Whole–part relationship (e.g., a Hotel has Rooms). Shown with a hollow diamond in UML.

Composition (“part-of”)

Whole-part relationship that is stronger than aggregation (e.g., Content is *part of* Post). Shown with a solid diamond in UML.

Inheritance (“is-a”)

Child class gets parent’s attributes/operations (e.g., Member *is a* RegisteredUser). Shown with a hollow arrow in UML.

Class::member notation

Use Class::member to show where an attribute or operation logically lives (e.g., Guest::doubleBeds).

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Advanced OO Terms

Encapsulation

Hiding how a system or component works internally and exposing only what users or other parts of the interface need to interact with it.

Coupling

An undesirable dependency between two or more objects/features.

Requirements and Features

Functional requirements

What the system must do; best expressed as stories of use.

Non-functional requirements (Constraints):

Qualities that shape design/implementation (e.g., performance, security, localization, usability targets). Two kinds: Technology and Quality-of-Service.

Features

A specific capability or function that a product provides to help users achieve something.

Stories of Use

Stories of use

Stories that describe user interaction, such as use cases, user stories, persona stories (persona-grounded), and scenarios (richer narratives).

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User story

A brief, one-sentence description of a user goal and the functionality needed to achieve it. Takes the form, “As a user, I want to [action] so that I can [goal].”

Persona story

Like a user story, but grounded in a specific, research-based persona. Takes the form, “[Persona] [Action] [Goal]”, for example, “Bob views available flights to make a booking.”

Use case

A structured description of system behavior from an actor’s perspective, including preconditions, flows, and postconditions. Bridges requirements and design with clear interaction steps.

Use case diagram

Overview of the relationships between actors and use cases.

Use case narrative

Detailed flows with Preconditions, Basic flow, Alternative flows, and Postconditions.

Scenario

A rich, narrative description of how a user or persona achieves a goal within a realistic context. Adds depth, emotion, and sequence to user interaction.

Epic

A large or complex story of use that combines several related user goals or interactions, and must be divided into smaller, clearer stories.

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Object-Oriented UI Design Tools

Unified Modeling Language (UML)

A common visual vocabulary for describing systems.

UML class diagram

The standard notation used to show classes/objects, their attributes, operations, and relationships.

Class naming (CamelCase)

Class names often use CamelCase without spaces (HotelCheckinView).

Design maps

Collaborative boards linking personas (who), stories/constraints divided into steps (what), and artifacts, like sketches and wireframes (how). Teams add questions, comments, and ideas.

Persona-weighted feature matrix

Used to rate features by importance to key personas.

Prioritization matrix

Used to balance importance vs effort/risk for implementation.

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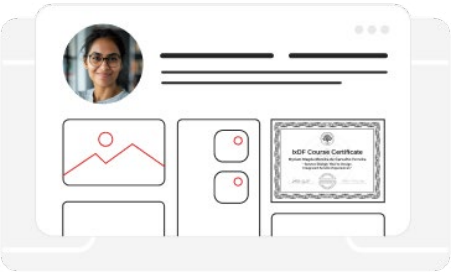

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